

Research Article

Canned Apple Fruit Freshness Detection Using Hybrid Convolutional Neural Network and Transfer Learning

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Fruits contain good nutrients such as protein and vitamins; therefore, fruits are usually used as complementary food ingredients. Apple fruit is one of the most important traditional table fruits in the temperate zone besides being the most commonly consumed fruit in the world. Apple's freshness is a decisive characteristic of consumer choice. Consumption may increase if apples of satisfactory freshness and quality are provided to the consumers. Reduced freshness of canned apple fruit results in significant spoilage during storage and distribution, affecting both agricultural profitability and consumer satisfaction. Many approaches have been presented to predict apple fruit freshness; however, most of these studies suffer from low accuracy due to poor image quality and the unavailability of sufficient data. This study aims to grade the freshness level of canned apple fruits by combining digital image processing and machine learning techniques. The proposed technique involves data collection of apple fruit, image segmentation, and preprocessing, creating the neural network model, training the network model, and testing for apple freshness prediction. A dataset is collected for apple fruit images for fresh, semifresh, and rotten classes, which is further augmented for more images. We extensively evaluated the performance of the proposed convolutional neural network-transfer learning (CNN-TL)-based technique. For performance evaluation, ResNet, GoogleNet, AlexNet, VGG, MobileNetV2, and InceptionV3 are also utilized due to their superior performance reported in the existing literature. The experimental results indicate the superior performance of the proposed approach with a 98% accuracy on the original dataset, which is better than other deep learning models. The model secures a 97% accuracy on the augmented dataset. Cross-validation results report a 97.71% accuracy using five folds. Furthermore, a comparison with existing state-of-the-art approaches shows that the proposed approach has better results for apple freshness classification.

Keywords: agriculture and human health; artificial intelligence; fruit freshness detection; image processing; neural network

Preserving the freshness of canned apple fruits is a major challenge in the food industry. Unlike fresh fruit, where visual cues such as color and texture can indicate quality, canned fruit is sealed, making it difficult to assess its internal condition without opening the packaging. This poses the risk that products with compromised freshness may go undetected during storage or distribution, leading to spoilage and economic losses [1–3]. The random sampling and manual controls used to check freshness are both time-consuming and sometimes unreliable. However, these methods often involve opening packed apples, which is both destructive and impractical for large-scale production environments.

Image processing plays an important role in the agriculture field. It has lots of applications to measure the quality of fruits and vegetables. There are different types of diseases found during the cultivation of crops that affect the freshness of the different fruits, especially apples. Numerous approaches can be used to identify disease in its early stages, and this will help to maintain the apple fruit freshness for a long time. The traditional way to identify the freshness of an apple fruit is to look at it with your naked eye, but it is not effective. However, with the help of the digital method, apple fruit detection can be accurate and save time [2–5].

Programmed recognition of food quality is a lot of fundamental due to the assimilability of cost specialists. According to the report of the World Health Organization (WHO), Pakistan spends roughly 120 billion dollars each

Food shortages are considered one of the biggest problems in our reality today. A large number of large food loads including both packed and unpacked foods are constantly wasted. One of the key features of this great food waste program is that there is no official framework to take appropriate action before the food is wasted. That is the reason for food shortages from one side of the planet to the other, especially in African countries. According to summaries of the Food and Agriculture Organization of the United Nations (FAO) based in Rome (Italy), the world's largest population is starving at 777 million people in 2016, 815 million in 2017, and an estimated 821 million people in 2018 for global food insecurity [12, 13]. In the years to come, food security, expansion, and forecasting the time of real food use will be a daunting task for professionals as the world's population grows and contacts the furthest reaches of progress with an enormous development rate. In the present circumstance, increasingly more food is expected to satisfy the requirements of that massively developed populace. Roughly 28% of our food items are squandered, and 7% of this wastage is because of a few basic issues connected with the most common way of bundling alongside food bundling hardware, bad quality of compression materials, and increased cost of composite materials [5]. The previously mentioned statistics show that products of the soil in pressing have turned into the most often utilized food components.

Different existing frameworks have been executed as a process for distinguishing the time of usability of stuffed fluid food as well as pressed nonfluid natural products, for example, chilly beverages, milk packs, pressed meat, and pressed salads. In other examinations, they are utilized to expand the timeframe of realistic usability of food infirmities [17]. These frameworks help the clients to accomplish a portion of their objectives, for example, keeping food from getting into a spoiled state. There are a lot of ways to deal with doing this errand as one of them utilizes a mobile phone

camera to anticipate and perceive food items naturally [18]. However, the function of measuring the properties of packed foods is not performed [12]. But, the real effort is to take images of packed foods to predict shelf life without making any impact on the food. Geometric features are found in [1] using image processing. Some of the methods such as accelerated shelf life testing (ASLT) were used to predict the shelf life of the food but used unpacked foods and fruits. The ASLT applies to various parameters. First and foremost, a dynamic demonstration of the various temperature reduction responses that can be obtained using zero-temperature kinetics, the first and second applications followed by other steps [19–23].

Distinguishing between food items even in poor condition is a difficult task, especially in a picture frame, and after a while, nutritional qualities are changed. For example, water levels, calcium, iron, magnesium, fat, and calories in certain foods change over time. This is a centralized food order problem as indicated and handled by [3]. Many works have been proposed to detect the condition of fruits such as their life, freshness level, class concerning nutrition, etc.; however, these approaches suffer from low accuracy.

This study proposed a framework for canned apple freshness detection. The proposed framework used a digital processing algorithm and a convolutional neural network (CNN) to identify and forecast the current condition of the packed apple. It recognizes the distinctive characteristics of that natural apple products like “fresh”, “semifresh”, and “rotten.”

Computer vision is widely used for evaluating food quality as per its surface properties such as coloration and surface texture. However, concerning canned products, these features are usually hidden by the product covers, thus making it difficult to use standard methods of visual inspection. To overcome this limitation, we adopt X-ray imaging to capture images within the canned apple fruits in order to determine their internal structure and likelihood of spoilage. These X-ray images are then passed through a computer vision algorithm working on CNN to predict the freshness of the fruits. This means that while X-ray imaging is used to acquire the data, computer imaging is used to analyze the freshness of the canned fruits.

- A canned apple freshness detection framework is prepared that utilizes a custom-designed (CNN) with transfer learning (CNN-TL). The proposed approach aims to detect the freshness of canned apple fruit.
- A self-created dataset of apple fruit images is utilized that takes pictures from within the container with different conceivable positions using an X-ray luggage scanner.
- Evaluation of the proposed framework in comparison with well-known models such as ResNet, GoogleNet, AlexNet, VGG, MobileNetV2, and InceptionV3 is done.

The main objective of this investigation is to present a technique that uses computer vision to assess the color characteristics of a large sample of apple fruits and then

build a model that predicts how long the fruits will last. Although the study focuses on apples, the methodology may be applied to any climacteric fruit that exhibits discernible color changes as it ripens. In essence, each fruit's features will be recorded and analyzed by the camera, which will then connect these features to the created model to produce accurate shelf-life predictions for that particular fruit variety.

Essentially, rather than making distribution decisions based on current features, this research proposes a model that allows current features to anticipate future features, which are then utilized to make distribution decisions. Finally, the proposed method would be beneficial because, unlike the existing fruit grading technique, it allows for an objective, accurate, and tangible shelf-life prediction system. This study offers an experimental method to go through, as well as the statistical analysis that follows, in order to create a predictive model of color attributes versus shelf life. Apples are used in the study as a proof of concept for the technique.

There are two experiments in this study: the first obtains images of apple fruit samples and then measures the features destructively. The second experiment's goal is to determine the end point of shelf life by correlating color representations with size content. The color of many apple samples is recorded on the first day that their sugar content reaches or exceeds 23%, and the color range is then set as the end color of the second experiment. The subsequent trial includes catching images of fruit products every day until it is rotten. The shading highlights are separated consistently utilizing computer vision. For factual examination, each natural product is plotted to quantify shading as an element of time. Posthandling then happens with various portrayals of shading being determined (utilizing existing computer vision calculations). Then, at that point, for each natural product, the timeframe of realistic usability is determined at a given color or worth. This is rehashed for each chosen tone and each organic product.

Section 2 discusses works related to this study, followed by the presentation of the proposed approach in Section 3. Results are discussed in Section 4, while the conclusion is given in Section 5.

2. Related Work

In this segment, we originally characterized a portion of the fundamental works connected with food misfortunes, fresh and spoiled states, date marking of food, expiration dates, etc. We likewise examined what factors add to food loss, for example, reasons for food loss. Food waste in various product categories is also discussed. Finally, we look at how to evaluate images based on pixels, as well as a neural network-based framework to predict the freshness of food.

The reason for this literature review is to give adequate information on DIP and its application in the food and farming industry nowadays. Furthermore, this section demonstrates examination loopholes and recent concerns related to food age determination. Then, the study examines various statistics related to food quality expectations. The biogenic amines index (BAI) is one of many techniques used to predict the freshness of meat. This is something that

involves testing for meat burns. The electrical nose is enhanced by the results obtained from the BAI by dispersive fluid micro-extraction gas chromatography. Electrochemical sensors with artificial brain networks are used to display E-nose. It is eliminated due to the nasal electronic effects sensor exhibits can be utilized to acquire solid outcomes quickly and for a somewhat minimal price of meat newness however, the business utilization of this sort of gadget is anything but a whimsical thought [24].

Computer vision systems face significant issues in estimating shape. This is a result of the consistency of natural products. Each natural product has its unique shape, and it is difficult to define the nature of the “ideal” natural product. To plan the shape, for the most part, there should be several classes and levels of recognition of what an object falls into. The manufacturer must create the right shape and then consider the resistance of the variety, extracting any organic product without that durability [17, 25].

In computer vision, an entity is analyzed by taking a picture of it and extracting features and characteristics for later processing (characterization, discrimination, model building, and so forth). Throughout recent years, computer vision has progressively been used in the agriculture industry. Normal utilization of computer vision frameworks in rural business includes programmed arranging, reviewing, and assessment of the nature of organic products, vegetables, and meat. Computer vision frameworks showed great results as they can be faster and more reliable than human grading experts. Computer vision systems have the advantage of being able to detect light in the ultraviolet and infrared spectrums, which humans cannot [26].

Computer vision requires three elements: a light source, an object to be blown, and a receiver to identify light. Lighting should be controlled to decrease downstream separating, upgrade the qualification of something interesting from the foundation, and reduce meditation. There are two different types of lighting: Front light is used to investigate the actual surface of a natural product (e.g., shading tests or evidence of imperfections). Backlighting is used to check the shape and size of an object. Front light feature detection and background light detection are common characteristics [27]. The use of all imaginable front lights or backlights depends on the business and is often used for various pieces of research. One remarkable method of controlling similar and recurring lighting conditions was eliminated by Leemans. The idea was to use a circular and clear cylinder with white paint to create a light mirror. This is considered in any case, lighting to fill the entire cylinder as opposed to learning to identify light sources [28, 29].

Size assessment is one of the primary elements to recognize business classifications of products. For push makers, size is the main trademark. “Size, which is the primary boundary related to quality, has been assessed utilizing machine vision by estimating either region, border, or on the other hand measurement.” To test size, there are two common strategies [30]. The main method is in the pixel area, which includes collecting each pixel as one or another location with an object and object (front area) or base. The following process involves considering past knowledge to

define a category (e.g. shading changes, limitations, location changes). This type of experiment was previously performed by Blasco in size experiments, and the image was considered a paired object, the main being a plant-based diet [31]. Another novel classification system is presented in [32]. Instead of separating the image, the authors use the split to see the object from the base. This was aided by the use of a blue base (as they liked the arches of red and white pomegranate) to make the front (object) and the base look more functional and showed promising results.

The study by ElMasry et al. [33] presented an approach to determine the type of apple using a VGGNet network. First, a dataset of fruit images is used to train the VGGNet architecture. The YOLOv3 model is then loaded with weights that have already been trained. The apple freshness assessment model and the apple type detection model are the two separate models that make up the system. Preprocessing is done, and then the VGGNet model is used to predict the type of apple. The image is then run through the YOLOv3 model to ascertain its freshness.

Steinbrener et al. used real-world images with complex disturbances to investigate the detection and categorization of apple quality [34]. The work presented a novel CNN-based model intended to accomplish accurate and quick apple quality ratings. The suggested model showed remarkable performance through rigorous training and validation, reaching peak training and validation accuracies of 99% and 98.98% at the 2590th and 3000th epochs, respectively. The suggested model's overall accuracy was 95.33% when evaluated on a separate dataset with 300 apples. A comparative study showed that in terms of training accuracy, overall test accuracy, and training length, the suggested model performed better than the Google Inception v3 model and conventional imaging techniques.

Using photographs of both fresh and rotting oranges, apples, and bananas in the dataset, the authors presented a method for classifying fruit freshness in [35]. The model performance was assessed using a variety of techniques, such as CNN, Inception, Xception, K-nearest neighbor (KNN), support vector machine (SVM), random forest (RF), decision tree (DT), and Naive Bayes (NB). With an accuracy of 97.25%, the suggested method with the Xception algorithm yielded the highest accuracy, ahead of the Inception algorithm at 95.55% accuracy. CNN scored 77.17%, RF 77.17%, SVM 75.87%, KNN 61.3%, NB 59.38%, and DT showed a 44.29% accuracy. The study by Cubero et al. [26] examined three apple diseases: apple scab, apple rot, and apple blotch. It uses K-mean clustering for image segmentation and extracts features like global color histogram, color coherence vector, local binary pattern, and complete local binary pattern. The authors reported 93% classification accuracy.

In another research by Chaudhry et al. [36], it was claimed that image sensors made from complementary metal-oxide semiconductors (CMOS) are essential for the accurate classification of products in agriculture. Fresh/rotten bananas, fresh/rotten oranges, and fresh/rotten apples were the three fruit groups that were the focus of the study's image categorization technique. To do this, the authors used over 30,000 RGB photos of the six different

The present research provides an in-depth evaluation of several image processing and computer vision approaches and techniques. Regarding the large number of methods proposed, their analysis is often context-specific. It is decided that more research projects are required to close this gap. This study provides a framework to determine the freshness of apple fruit in particular; however, it can be used for other fruits and vegetables as well.

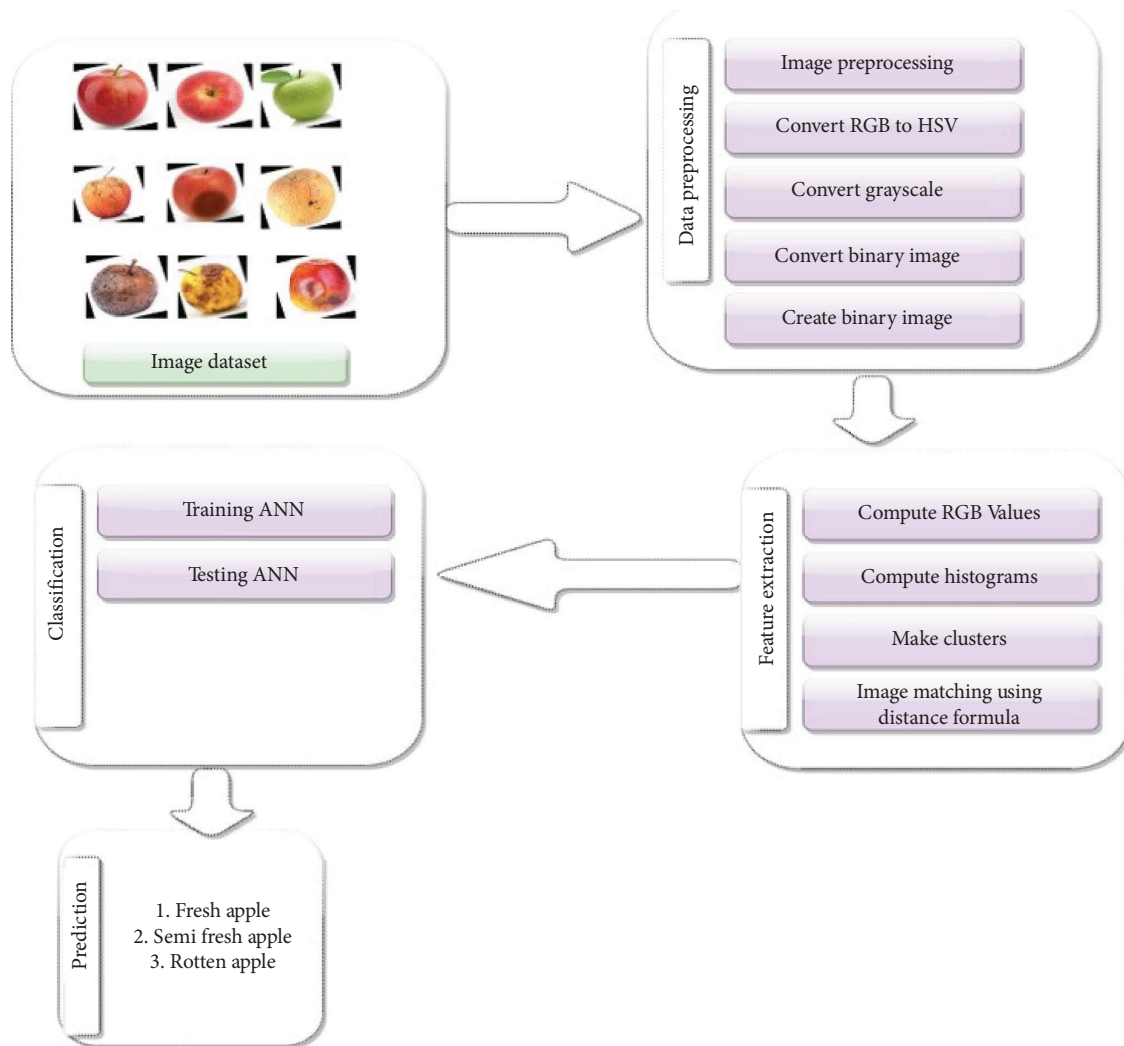


FIGURE 1: Proposed framework for the analysis of apple fruit freshness.

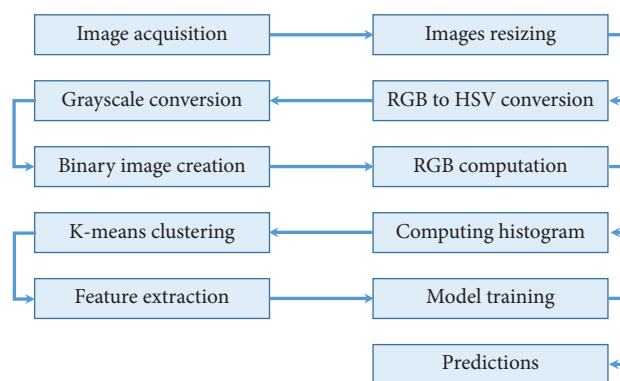


FIGURE 2: Packed apple fruit freshness prediction algorithm.

3.5. Build Datasets. For experimentation, exactly the packed apple dataset could not be found. Therefore, for experimentation and performance analysis, a total of 756 raw images of apples are collected on three different days, as shown in Table 1. Variations in time, angle, and light intensity were purposefully added during the image-

gathering process to improve the recognition model's accuracy and robustness. Using the same mobile camera (13 megapixels, $F\text{-Stop} = f/2$, exposure time = $1/50$ s, ISO speed = 151, focal length = 4 mm, Redmi Note 4X, China), 345 original photos with a resolution of 3120×4160 pixels were taken.

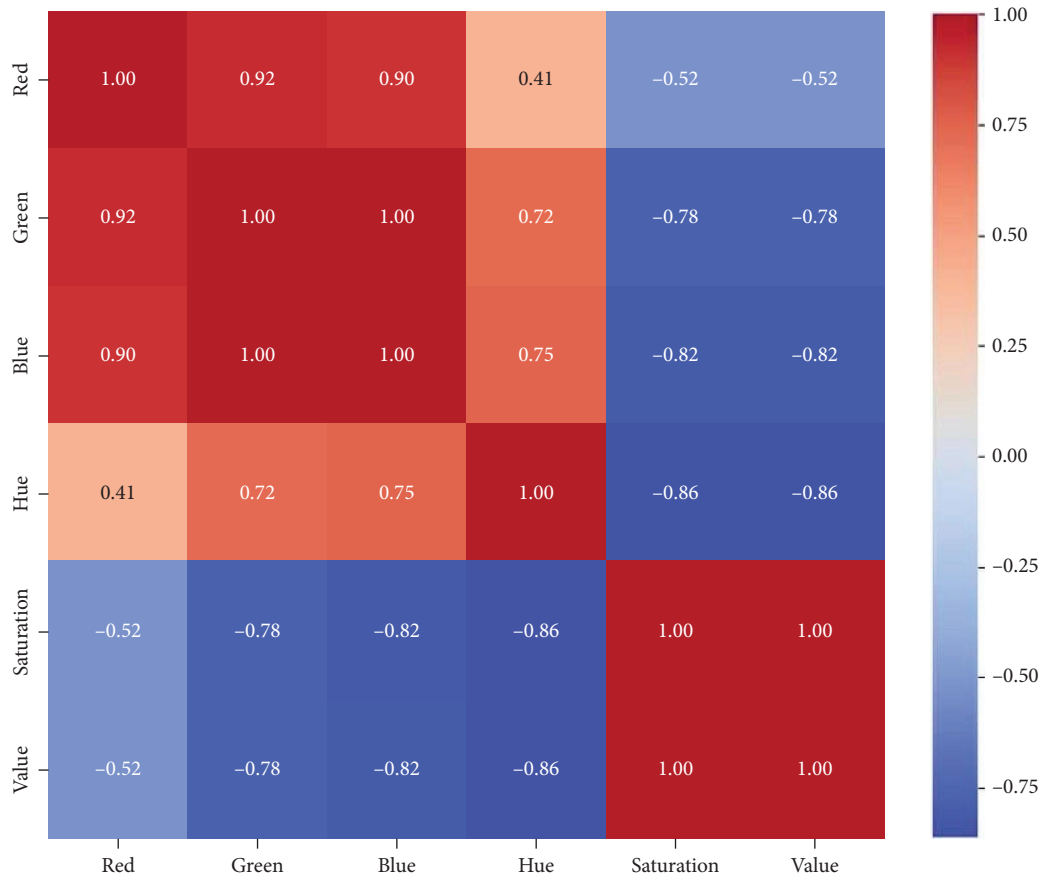


FIGURE 3: Correlation between color features and freshness categories.

TABLE 1: The statistical details of the collected apple images.

S. no.	No. of days old	Class	No. of images
1	20–30	Fresh	420
2	30–40	Semi fresh	210
3	> 48	Rotten	126

As there are no specific existing apple fruit freshness datasets available, this research work comprehends the detailed analysis for the collection of the data. For this study, there are two different datasets collected and used for experiments. In the first apple fruit dataset, there were a total of 756 raw images of apples on three different days, being captured. To capture the apple fruit images with good quality, we have used the mobile camera (50 mega-pixels, $F\text{-Stop} = f/2$). We have different images for different classes of apple fruit.

The following criteria are used to define quantitative differences between varied freshness states: These criteria depend on the visual properties, familiar to the food industry. The main features looked at in this occasion are color, surface features, and signs of spoilage.

- **Fresh:** fruits in this category must be shiny, possess high skin quality, and have no spots or signs of flesh softening. These apples remain hard and intact in color which shows that these are the best apples for consumption.

- **Semifresh:** semifresh apples will show very little sign of rotting that they depreciate in color, get slightly woody, or have slight skin abnormality. However, they are still in some ways edible, though the quality is lowering, which limits their attractiveness.
- **Rotten:** wasted apples are much more visibly impaired than semiwasted apples; they are substantially darker or browner, softer, or pulpy, or noticeably moldy. These are bad apples and warrant a potential health risk should they be consumed.

3.5.1. Data Augmentation. Data augmentation approaches have been used to overcome the problem of restricted training dataset size. When there are inadequate data, data augmentation offers a straightforward and practical way to improve model resilience, increase dataset diversity, and reduce overfitting problems. Data augmentation lessens the model's dependence on particular characteristics or features by modifying training samples. By producing more useful

data from the current dataset, this method enhances the robustness and performance of the model.

- **Fresh:** Initially, there were 420 images in this category, and after augmentation, the number of samples increased to 960.
- **Semifresh:** The original 210 images were augmented to a total of 595.
- **Rotten:** The 126 images in this category were augmented to 575.

3.5.2. Process of Augmentation. The data augmentation techniques applied in this study involved multiple image transformations such as flipping, rotating, and adjusting brightness and contrast. These procedures were implemented to create variations in the dataset while maintaining the core semantic features of the images (i.e., ensuring that the freshness indicators such as color and texture were not altered in a way that misrepresents the actual fruit condition). The augmentation was done to increase the diversity of training samples without distorting the key characteristics used for freshness classification.

The augmentation techniques used aimed to simulate natural variability in real-world apple images, such as changes in lighting, angle, and texture, without distorting the core features that are critical for freshness classification. The key augmentation methods applied include the following:

- **Rotation:** Images were rotated at angles of 60°, 120°, 180°, 240°, and 300°, producing five variations per original image.
- **Flipping:** Horizontal flips were applied to create more diverse training samples.
- **Brightness and contrast adjustment:** Brightness was increased by 1.5, and darkness was reduced to 0.9.
- **Noise injection:** Salt-and-pepper noise and Gaussian noise were added to prevent overfitting and to mimic real-world image noise.

These transformations were applied to all categories (fresh, semifresh, rotten) to ensure each class was equally represented after augmentation.

The original photographs were downsized to quarter-sized images, which produced a resolution of 780 × 1040 pixels, to speed up the data augmentation process. Each scaled-down image in this study underwent a variety of augmentation procedures intending to preserve image semantics while increasing dataset diversity. These procedures, such as flipping, rotating, brightness adjustment, and noise, ensured that key features related to freshness were not inappropriately altered, thereby maintaining the integrity of the datasets for accurate classification. Adding salt and pepper noise, boosting Gaussian noise, flipping, rotating, adjusting brightness, and adjusting darkness were among the techniques used. By diversifying the input image characteristics, an increase in Gaussian noise effectively prevented overfitting, with a salt-and-pepper noise density of 0.3 being used. Brightness and darkness were modified by 1.5 and 0.9,

respectively, after horizontal flips were applied. Each sub-image underwent rotational manipulation to produce five more subimages at angles of 60°, 120°, 180°, 240°, and 300°. Thus, using these augmentation methods, 2130 apple sub-images in total were produced. Table 2 provides specifics on the dataset's postaugmentation makeup.

The necessity of data augmentation arises from the limited original dataset size, particularly in the Semi Fresh and Rotten categories, which had fewer images. This limitation can lead to model bias if one class is underrepresented. Augmentation helps in the following:

- **Increasing the dataset size:** The original dataset consisted of only 756 images, which was insufficient for training a robust model. Augmentation expanded this to 2130 images, which is a more suitable size for model training.
- **Improving model generalization:** By introducing variability into the training set, the model becomes more adaptable and less likely to overfit specific features of the original images.
- **Balancing class distribution:** Augmentation mitigates the issue of imbalanced classes, ensuring that the model does not become biased toward the Fresh category, which had a higher number of original images.

The balancing through augmentation ensures that the model receives sufficient examples from all freshness stages, improving the overall accuracy and fairness of the predictions.

Two separate sets of the dataset were created to train and evaluate the suggested model. The augmented and original dataset makes a total of 2130 apple subimages in the dataset designated for training. Both training and testing sets were independent and chosen at random from the dataset. These images were then stored on the computer and used in the data analysis procedure for the categorization of apples. Figure 4 shows sample images from different classes of apples from various angles.

3.6. Preprocessing of Datasets. To prepare image data for computer analysis, preprocessing is an essential step in image processing. It is crucial for getting images ready for testing and comes before the major processing stage. Preprocessing the dataset is an important next step after compiling the packed apple dataset. To improve the sharpness, clarity, texture, and masking of the packed apple picture data, image preprocessing is utilized. This includes activities like eliminating low-frequency image components, decreasing noise, and adding masking. This preliminary stage has a beneficial effect on the model's accuracy and is essential in optimizing the dataset for additional processing.

After the images were moved to a computer, additional processing was possible, allowing grayscale values to be gathered for the categorization of apples. The color of apples has a significant impact on customer perception, making it an important factor. Apple color changes as a function of freshness level, while stored at 35°C [32]. At higher storage

TABLE 2: Dataset composition after image augmentation on the training set.

Class	Size of original datasets	Size of augmented datasets
Fresh	420	960
Semifresh	210	595
Rotten	126	575

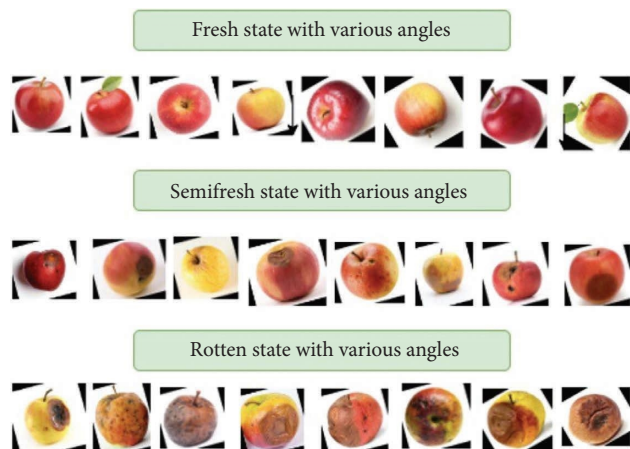


FIGURE 4: Samples from augmented images indicating various freshness levels.

temperatures, such as 35°C, apples undergo accelerated metabolic and enzymatic activities, leading to changes in color, browning, and overall freshness loss. These changes in color, from vibrant red to brown or darkening, serve as an indicator of freshness decline. The basis for this observation is supported by studies in postharvest physiology and fruit ripening behavior under temperature stress. Apples that are semifresh may have brown patches, while fresh apples usually have red coloration. The RGB values exhibit these color shifts. Freshness qualities are grouped into second and third categories based on these color alterations. In addition, apples tend to lose nutrients and become smaller as they become less fresh.

Preprocessing techniques were applied to images of apple fruits in order to accurately extract apple features. These actions included creating binary pictures to standardize the intensity of each image, converting from RGB to HSV, and converting from grayscale. These steps were done to shorten the framework's computation time. To speed up prediction results, the whole dataset was downsized to 100 × 100 pixels size. In order to identify genuine spots, techniques were also used to transform RGB photos to grayscale images and create binary images from the original images.

3.6.1. Converting HSI to Gray Scale Image. Using the Otsu approach, the main goal of this process is to produce binary images. This method is based on locating the histogram's minimum point that falls between two classes while accounting for interclass variance.

3.6.2. Image Filtering. An average filter was implemented in this process to remove Gaussian noise. The average filter computes the mean (average) of the grayscale in a rectangular filter window surrounding each level. It will remove the noise permanently.

3.6.3. Binary Image. For the sake of keeping a uniform dataset size, all photos were downsized to 100 × 100 pixels during this phase. Each image was then transformed from RGB to HSV color space. The HSV color space is used in image processing because it makes light intensity simpler and makes color identification easier, especially when it comes to differentiating between apple fruits of different maturation degrees. After that, in order to save computing expenses, the image was changed to grayscale. It was then turned again into a binary image so that morphological operations could be applied, which are crucial for locating holes in the images.

3.7. Feature Extraction. Feature extraction is the process of using particular techniques to determine an image's key attributes. In this study, color and texture, two important characteristics, are used to evaluate the freshness of packed apple fruits. Features like as color, texture, and form are frequently used to assess the freshness of apple fruit; however, the success of these techniques mostly depends on feature extraction.

3.7.1. Color Pattern Based on Histogram. It is frequently recognized that fruit and vegetable color plays a critical role in quality recognition systems and customer satisfaction. Figure 5 shows the histogram statistics for apple. The RGB color model is utilized in this study to represent color. Three discrete colors, red, green, and blue, defined in a 3 × 3 matrix make up an RGB image. Every image in the database has its RGB value computed and saved for comparison with query images.

$$H = (\cos)^{-1} = \frac{(1/2)\sqrt{[(R-G) + (R-B)]}}{\sqrt{(R-G)^2 + (R-B)(G-B)}} \quad (1)$$

3.7.2. Morphology Processing. The analysis of structure is referred to as morphology. Mathematical morphology is used in medical image processing to recognize and extract significant picture features from the image based on its shape attributes. Morphological operations are logical transformations that compare the neighborhood of a pixel with a structural element, which is a predefined pattern [37].

Erosion and dilation are the two basic morphological processes. Dilation gives items the ability to enlarge, possibly bridging tiny spaces and joining disparate elements. Conversely, erosion causes objects to contract by weakening their borders. More complex transformations can be produced by combining these two basic processes, erosion and dilation. Morphological filtering frequently uses opening and shutting processes. The same structural member is used

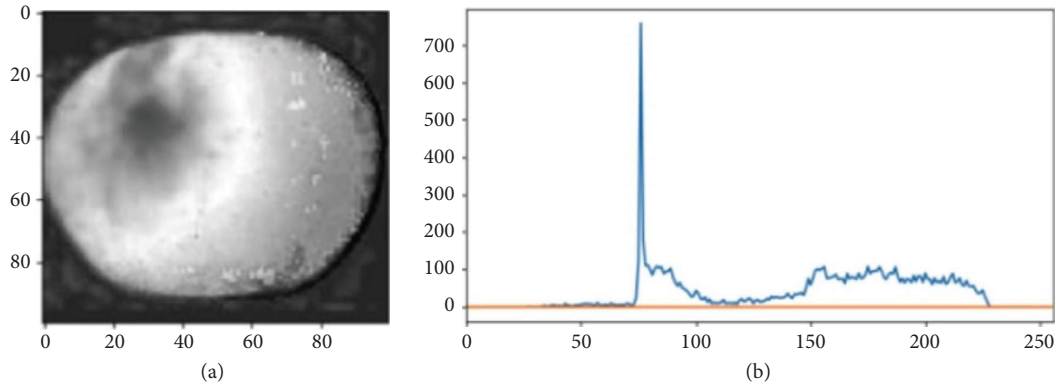


FIGURE 5: Histogram statistics.

for both dilatation and erosion during opening. On the other hand, closing involves dilation and erosion, essentially carrying out a reverse-opening process using the same structural element [38, 50].

K-means clustering uses a clustering technique to perform feature extraction. It is an unsupervised learning algorithm, we used unlabeled data to define the different states of apple. K-Means divides the object into clusters that share the similarity and dissimilarity related to the object. A typical K-Means clustering algorithm is given in Algorithm 1.

There is a centroid point in an object, which is computed by minimizing the following objectives:

$$K = \sqrt{(X_o - X_C)^2 + (Y_o - Y_C)^2}, \quad (2)$$

where K denotes no of clusters, o is an observed value, and C is the centroid value; we performed this calculation to find the minimum distance of each cluster.

$$U = (X_o + X_n) \div 2, Y_o + Y_n \div 2, \quad (3)$$

where U is the new centroid value, X_o is an old cluster coordinate, and X_n is the new cluster coordinate, similar to Y_o and Y_n .

3.8. K-Means Features Extraction. The main aim of K-means is to enhance the segmented image, so it will help an accurate classification of an apple if there is some defect.

In the proposed approach, a square Euclidean distance is used to measure K-Means clustering. The algorithm classified and predicted their quality. The modified K-Means with square Euclidean distance for clustering for packed apple fruit freshness prediction is given in Algorithm 2.

3.9. Working of Proposed Approach. Image segmentation using the K-Means algorithm is very useful. The basic aim of image segmentation is to clear the background and object. Figure 6 shows the images without segmentation. Image segmentation helps in obtaining those areas that are affected, as shown in Figure 6(b).

The main goal of the proposed approach is to segment color automatically using K-Means clustering and $L^*a^*b^*$ color space. The proposed framework works in the following steps:

- Import the image of the faulty fruits.
- Transform the RGB image into Lab^* color space. Due to the presence of two chromaticity layers in the “ a^* ” and “ b^* ” channels and a luminosity layer in the “ L^* ” channel, the Lab^* color space was selected. Because all color information is contained within the “ a^* ” and “ b^* ” layers, this selection is computationally efficient.
- To identify colors, use K-Means clustering in the “ ab ” space. The difference between two colors is measured using the Euclidean distance metric.
- Using the K-Means clustering results, give each pixel in the image a label. An index for each cluster is assigned to each pixel.
- Create segmented pictures by using color segmentation as a basis. Using pixel labels to represent the amount of clusters found, the image is divided into many images.

Figure 7 shows the graphical representation of the defected clusters identification from a given input defected apple image.

The algorithmic index of each cluster containing the faulty fruit piece is calculated, as K-Means may not always produce the same cluster index number. Alternatively, the cluster center value, which comprises the average value of “ a^* ” and “ b^* ” for each cluster, can be used to achieve this. Several images of damaged apples in packaging are displayed in Figure 8.

When the K-Means clustering algorithm is applied to the packed apple fruit for clustering, the resulting clustering is shown in Figure 9.

3.10. Selecting Appropriate Hyperparameters. In order to enhance the performance of the deep learning model, hyperparameter tuning was carried out using different

1. Select K points as initial centroids
2. Repeat
3. Form K clusters by assigning each point to its closest centroid
4. Recomputed the centroid of each cluster until the centroid does not change

ALGORITHM 1: K-Means clustering.

1. Apply K-means to sort out the colors of apples using the Euclidean distance method.
2. Label each pixel in the image using the results of K-means.
3. Create color-segmented pictures from an input image.
4. The median filter is then applied to the segmented image.
5. Now the images are the input values for the classification procedure.

ALGORITHM 2: Modified K-means clustering.

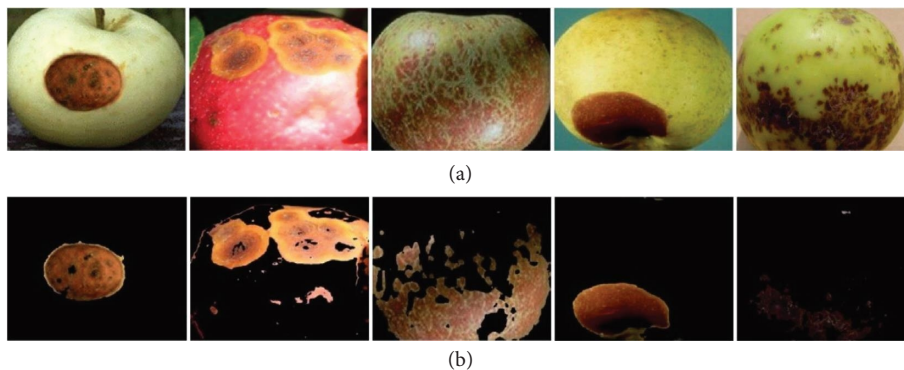


FIGURE 6: Image segmentation results: (a) before segmentation and (b) after segmentation.

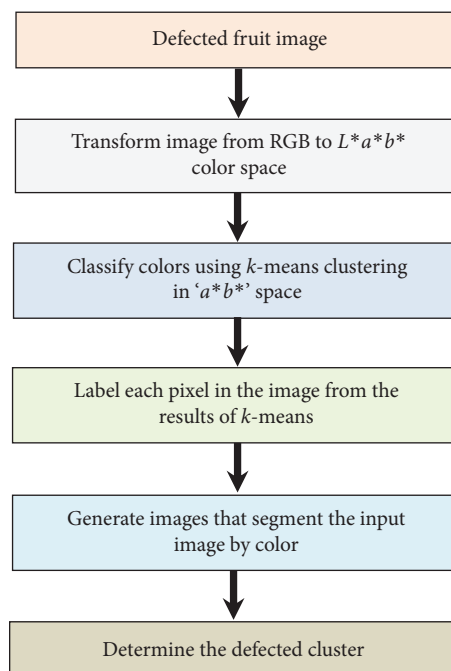


FIGURE 7: Discovery process of defected clusters in the defected apple fruit image.

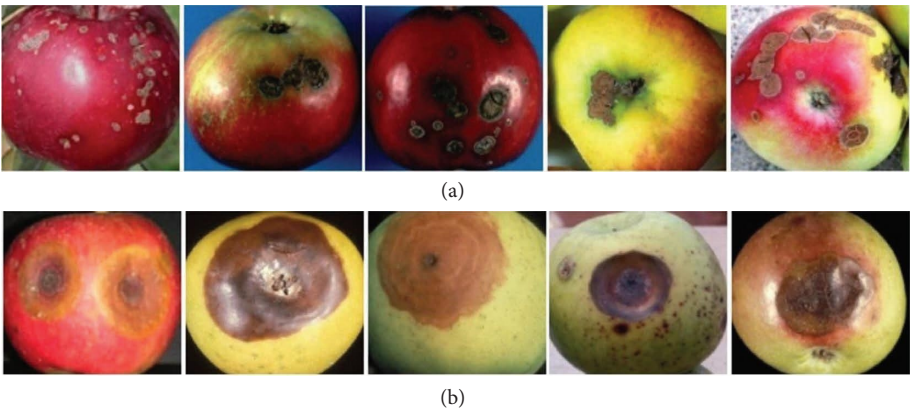


FIGURE 8: Samples of some defective packed apple images.

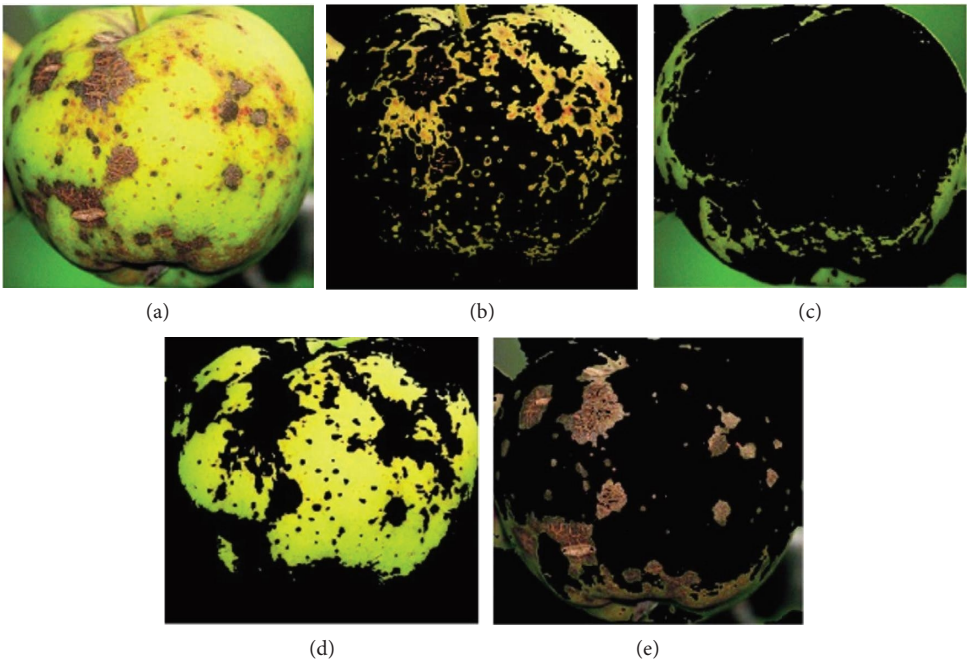


FIGURE 9: When apple scab disease-affected fruit was subjected to K-means clustering, four clusters were produced. (a) The original image of the infected fruit. (b) The first cluster. (c) The second cluster. (d) The third cluster. (e) The fourth cluster.

learning rates and batch sizes. These hyperparameters were chosen experimentally, with different values tested to determine their effect on model accuracy and training time. Table 3 summarizes the hyperparameter settings and their corresponding results during the tuning process.

It is observed that a learning rate of 0.001 and a batch size of 32 resulted in the highest accuracy (85%) and the shortest training time (100 s). The accuracy decreased when batch sizes were increased to 64, possibly due to instability in training, as larger batches may have led to slower convergence. Additionally, the accuracy with a learning rate of 0.01 was generally lower, indicating that a higher learning rate might have caused the model to overshoot optimal minima during training. These results led to the selection of learning rate = 0.001 and batch size = 32 as the best combination for model performance.

TABLE 3: Hyperparamter tuning for CNN model.

Learning rate	Batch size	Accuracy (%)	Training time (seconds)
0.0001	16	72	130
0.0001	32	75	125
0.0001	64	70	140
0.001	16	80	110
0.001	32	85	100
0.001	64	78	120
0.01	16	68	145
0.01	32	74	130
0.01	64	75	140

For the activation functions, the rectified linear unit (ReLU) was used because it reduces the problem of “vanishing gradient.” This was useful to speed up the training by

bringing nonlinearity into the training while keeping the gradients' tame. Transfer learning enabled the model to incorporate certain prior information in a way that enhanced its performance and sped up training when data were scarce. Furthermore, functions like image rotation, flipping, and brightness adjustments were also applied to augment the training set. This also assisted the model to perform better in its generalization, plus avoiding the vice versa of overfitting.

3.11. Classification Using CNN and Transfer Learning. In the suggested approach, the classification of the apple fruit freshness has been performed using the well-known and adopted transfer learning technique. Further, in the suggested transfer learning this study involved the adoption of the AlexNet architecture. In the subsequent paragraph, a detailed discussion of the architecture is presented. In AlexNet architecture, there are a total of eight layers. These layers are decomposed into two classes: The first, there are five layers in the AlexNet, which are called convolutional layers; and layers in the second class are fully linked layers. These two classes of convolutional layers and linked layers form the architecture of the AlexNet.

AlexNet is particularly selected due to its deeper architecture. It has eight layers, which makes it effective in extracting features compared to other models. It particularly works well with color images. In addition, the ReLU function in AlexNet does not limit the output and there is less loss of features. Since many perceptrons are not active in the AlexNet, it improves the speed of model training.

CNN was chosen due to its unparalleled ability to process and extract key features in high-dimensional image data, making it indispensable for detecting object quality (e.g., freshness detection in produce, or defect detection in manufactured products) [51]. For instance:

- Hierarchical feature learning: CNN automatically learns hierarchical patterns (edges, textures, and shapes), which are essential for tasks requiring precise differentiation, such as identifying signs of deterioration or detecting abnormalities in images.
- Robustness to variations: CNN handles challenges like variations in object orientation, scaling, and illumination, common issues in real-world image capture.
- Data efficiency: Parameter sharing and local connectivity ensure CNN can efficiently train on relatively constrained datasets, which is often critical in industrial or agricultural scenarios.

From the dataset of apple fruit images, images are read as input to the first layer of the AlexNet, and the size of all the images is converted to 256×256 . This way, the output of the first layer is used as an input for the next layer, using 256 convolutional filters using a resolution of 5×5 . In the next layers, the third and fourth layers of the model, and 384 convolutional filters were applied, by resolution of 3×3 . After the task of the third and fourth layers is finished, layer 5 comes into action and here using 3×3 resolution total of 256 kernels were applied. In convolutional layers, maximum pooling with the resolution of 2×2 was used using the

scheme of batch regularization. Further, the significance of the selection of the activation function plays a vital role in enhancing the accuracy of the classification task. Also, the selection of the appropriate function must ensure that the value of the gradient function must converge rapidly and that the infinity value of the selected activation function should not return a value of 0.

For this study, the ReLU function is used as an activation function, which very efficiently tackles the issue of the gradient, as discussed above. ReLU function has been selected as an activation function as it is the simplest and needs no complex processing for computing its required value and consequently, the chosen model may get its training in a short duration. Transfer learning is a very famous deep learning approach, which involves the reuse of the pre-trained model to enhance the accuracy of the model in use.

Several hyperparameters were chosen experimentally in the process of fine-tuning the model. The learning rate of 0.001 has been chosen out of trial values of learning rate 0.0001–0.01. Compared to other values, the $\tau = 0.1$ value rendered the best compromise of achieving rapid convergence as well as checking with established updates by the different epochs. We also tried 16, 32, and 64 as batch size where we found that 32 was a good option because it gives the best result with less time required for training a model. For the activation functions, the ReLU was used because it reduces the problem of “vanishing gradient.” This was useful to speed up the training by bringing nonlinearity into the training while keeping the gradients' tame. Transfer learning enabled the model to incorporate certain prior information in a way that enhanced its performance and sped up training when data were scarce. Furthermore, functions like image rotation, flipping, and brightness adjustments were also applied to augment the training set. This also assisted the model to perform better in its generalization, plus avoiding the vice versa of overfitting.

4. Results

This study performs experiments with a self-collected dataset. As the dataset size was small, data augmentation was performed to increase the number of training samples. To avoid data leakage problems, augmentation is only used for training data and the testing phase uses the originally collected images. Finally, 1800 apple subimages were obtained. These generated images were added to the dataset composition. The dataset is divided into two sets for training and testing the proposed model. The training and testing sets were independent and randomly sampled from the originally collected images in a ratio of 0.8 to 0.2. The training set was then augmented to increase its size for proper training of the proposed model.

The proposed CNN-TL model was trained on an augmented dataset, requiring approximately six hours on an NVIDIA GTX 1080 GPU. For deployment, the model achieves a processing time of under 100 milliseconds per image, making it suitable for near-real-time applications. The current model can be implemented in systems with moderate latency, but real time may require additional

optimization techniques such as pruning and quantization for the model. These techniques besides cloud-based deployment ability would enhance the ability of the model to operate with resource-constrained techniques, thus allowing applicability in real-time systems.

4.1. Win-Loss Ratios. In Table 4, win-loss ratios are depicted, which are computed using the defective images of the apple fruit found in the dataset.

4.2. Performance Analysis of Proposed Approach on Augmented Dataset. In this experiment, we selected the same number of images of the apple fruits from the testing split of the dataset and we examined how the proposed approach classifies each image of the apple fruit from various classes. Experimental results show that the suggested model prediction of the rotten class apple was high, and it achieved a prediction accuracy of 94.34%. Table 5 shows the classification results of the proposed approach for the “fresh,” “semifresh,” and “rotten” classes of apple fruit, indicating that the “semifresh” has a lower classification accuracy of 90.47%. The lower accuracy of the “semifresh” class is due to varying marks of the decomposition process found on apples. The nature and size of the decomposition are different leading to clusters of different sizes and numbers, thereby making it difficult for the model to discriminate between the “rotten” and “semifresh” classes. For the “fresh” class, the accuracy is 96.15% and the model obtains a mean accuracy of 93.95% accuracy.

4.3. Performance Analysis of Proposed Approach on Original Dataset. In addition to the augmented dataset, experiments are also carried out using the original dataset. Table 6 shows the confusion matrix for the proposed approach on the original dataset, as well as, the values for different performance metrics. It is observed that using only the original dataset, the accuracy of the proposed CNN-TL model is high with an accuracy score of 0.957, 0.94, and 0.96 for the “fresh,” “semifresh,” and “rotten” classes, respectively. Overall, a mean accuracy score of 0.96 is obtained with the original dataset while precision, recall, and *F1* scores are 0.94, 0.95, and 0.98. Close agreement between these metrics indicates a good fit of the model and shows that the model does not exhibit overfitting.

The proposed CNN-TL model shows 18 wrong predictions for the “fresh” class, while 402 predictions are correct leading to an accuracy score of 0.957. For the “semifresh” class, performance is even better with only 6 wrong predictions. While for the “rotten” class, the model made 6 wrong predictions out of 126 total predictions.

It is observed that the misclassification error for the “semifresh” class is higher than for the two other classes. As stated previously, the “semifresh” class has mild browning and similar other minor spots on apples. This might be similar to the “fresh” class samples for some samples, thereby leading to misclassification. Similarly, the medium-sized rotten and browning spots on the “semifresh” class might resemble those of “rotten” class for some samples, leading to

samples classified as “rotten.” Looking at the results given in Table 6, it can be observed that more samples from the “semifresh” are classified as the “fresh” and “rotten” classes compared to other classes, confirming this explanation.

4.4. Comparative Analysis With Deep Learning Models. For further exploration, different transfer learning models have been implemented in this study including ResNet, GoogleNet, AlexNet, VGG, MobileNetV2, and InceptionV3. These models are reported in existing studies for their better performance on image classification tasks and are adopted for performance comparison with the proposed CNN-TL model. Table 7 shows the comparative results of CNN-TL with other deep learning models implemented in this study. Results indicate a better performance of CNN-TL with a testing accuracy score of 0.96. Among the deep learning models, AlexNet and InceptionV3 models perform well, each with a testing accuracy of 0.95.

4.5. K-Fold Validation. K-fold cross-validation is a suitable approach to validate the performance of a model. The data are split into several smaller sets, and each subset is used to test the model’s performance, while $k - 1$ subsets are used to train the model. In this way, the model’s performance is analyzed on smaller sets of data to evaluate its generalizability. Results of fivefold cross-validation are presented in Table 8, indicating better performance of the proposed CNN-TL approach.

4.6. Performance Comparison. As depicted in Table 9, the proposed CNN-TL model is compared with existing models from the literature. The performance of the proposed CNN-TL architecture was evaluated using the accuracy of the original dataset. For comparative studies, the novel suggested work in this study outperforms the existing models and thus acquires the accuracy of 96% on the original dataset and 94% on the augmented dataset. This clearly shows the supremacy of the suggested CNN-TL model.

4.7. Limitations. This study proposed an efficient approach to detect the freshness of canned apple fruit. For better accuracy, a customized CNN model is used with K-Means feature extraction. Despite providing better results compared to existing approaches, it has the following limitations:

- The model achieves a processing time of under 100 milliseconds per image; however, for real-time applications, further improvements are needed.
- The approach involves data augmentation which although provides altered images concerning shape and color, though it does not provide the same results as it might on the original datasets. Further experiments are needed using a large dataset without data augmentation for further validation.
- Data resampling approaches, such as SMOTE and ADASYN, can be explored to analyze the difference in the model’s performance with data augmentation.

TABLE 4: Data samples for win and loss ratios.

Class name	Total images	Defected images	Win (%)	Loss (%)
Fresh	164	—	95.7	4.3
Semifresh	50	8	84	16
Rotten	164	7	95.7	4.3

TABLE 5: Results from testing dataset.

Apple state	No. of apples	Apple states correctly predicted (%)	
		No. of apple	% of apple
Fresh apple	84	80	95.23
Semifresh apple	42	38	90.47
Rotten apple	26	25	96.15
Average	152	60	93.95

TABLE 6: Performance analysis of the proposed approach on the original dataset.

Apple state	Predicted class			Performance metrics			
	Fresh	Semifresh	Rotten	Accuracy	Precision	Recall	F1 score
Fresh	402	12	6	0.957	0.93	0.94	0.98
Semifresh	6	204	0	0.971	0.94	0.96	0.99
Rotten	2	4	120	0.952	0.96	0.97	0.98
Average				0.96	0.94	0.95	0.98

TABLE 7: Comparative analysis of the proposed approach with existing deep learning approaches using augmented and original datasets.

Approach	Original dataset		Augmented dataset	
	Training	Testing	Training	Testing
ResNet	0.72	0.73	0.77	0.77
GoogLENET	0.93	0.92	0.93	0.92
AlexNet	0.96	0.95	0.95	0.93
VGG	0.94	0.95	0.94	0.93
MobileNetV2	0.87	0.88	0.88	0.87
InceptionV3	0.95	0.95	0.95	0.93
Proposed CNN-TL	0.98	0.96	0.97	0.94

TABLE 8: Validation results of the proposed CNN-TL.

Fold	Accuracy	Precision	Recall	F1 score
1st	97.35	97.41	96.26	97.31
2nd	97.34	97.53	97.57	97.41
3rd	96.56	97.85	97.97	97.91
4th	97.90	97.96	97.89	97.92
5th	97.89	97.61	96.97	97.50
Average	97.71	97.64	96.97	97.59

TABLE 9: Comparative analysis of the proposed approach with existing approaches using the original dataset.

Approach	Accuracy	Precision	Recall	F1 score
Miah et al. [22]	0.94	0.93	0.89	0.93
Mudaliar [23]	0.91	0.95	0.87	0.94
Miriti [30]	0.93	0.95	0.88	0.93
Valentino et al. [31]	0.91	0.93	0.87	0.91
Proposed approach	0.96	0.94	0.95	0.98

- The dataset used in this study is from an X-ray luggage scanner for Braeburn Apple only. Model performance with data from another luggage scanner for other apple varieties may produce very different results.

5. Conclusions and Future Work

This study endeavors to automate the classification of canned apple fruit using a CNN with transfer learning and K-means clustering. A detailed analysis of the various apple fruit freshness techniques has been performed to explore their major contributions and limitations. Based on the identified limitations, this study proposed a novel image processing and CNN-based approach to analyze and check the freshness of apple fruit using the collected dataset of the apple fruit. A dataset of packed apples is gathered using an X-ray luggage scanner. Experiments are performed using the original, as well as, the augmented dataset. Experimental results indicate a similar performance of the proposed CNN-TL model with 96% and 94% accuracy for both datasets, respectively. Finally, the comparative analysis of the proposed approach with the existing approaches available in the literature has been conducted showing better results of the proposed model.

Although the introduced model in this study has shown very promising results and achieved 96% accuracy on the original dataset, it still requires further improvements in accuracy. The proposed approach has the potential to use deep learning and the internet of things, which is the topic of many current papers in the field. We have used the proposed approach for apple fruit freshness classification; however, the described approach can be extended easily to classify other fruits including oranges, mangoes, peaches, etc.

In addition to the current performance evaluation of the CNN model, future work will involve applying visualization techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) to enhance model interpretability. Grad-CAM will enable us to visualize the regions of apple images that the model focuses on when predicting freshness. This analysis can help in answering questions like if the model is paying attention to a feature of an image like surface defects or a change of color that defines the freshness degradation. In addition, there is a possibility to assess when to conduct the feature map fine-tuning to obtain higher accuracy. Such interpretability techniques may be used for enhancing the performance and the interpretability of a model and thus present reasons for the model in real-life applications.

Data Availability Statement

The data are available on reasonable request from corresponding authors.

Conflicts of Interest

The authors declare no conflicts of interest.

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